

Multi-Material Printing

Standard Operating Procedure

Equipment Location: PawPrint Studio

Ingram Hall Makerspace
Ingram School of Engineering
College of Science and Engineering



Policy Reminders

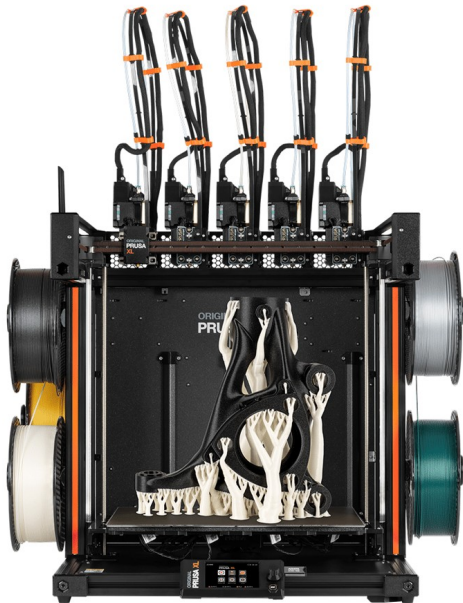
- Reserving a 3D Printer in FOM is not only required, it is impossible to print anything without doing so
- The following trainings are required for using the Prusa CoreOne or Prusa Mini printers
 - Fused Filament Fabrication Policy and Procedures Quiz (FOM)
 - LinkedIn Learning FFF Certificate
- In addition to the above trainings, the following trainings are also required for using the Prusa MMU or PrusaXL printers:
 - Multi-Material FFF Policy and Procedures Quiz (FOM)

Policy Reminder (cont.)

- Unless a printer is labeled “BYOF” the only allowable filament is PLA supplied by the Makerspace
 - There is a 30-minute minimum charge for all 3D Printer usage
- Printers labeled “BYOF” may not use any Makerspace filament and the user supplies their own
 - Any filament is allowed, provided it matches the capabilities of the printer
 - Ask a Makerspace staff member for questions regarding material compatibility
- Avoid touching moving parts while printing
- Be careful of the printer's hot end
- The Prusa CoreOne MMU is only to be loaded by an MST

Equipment Overview

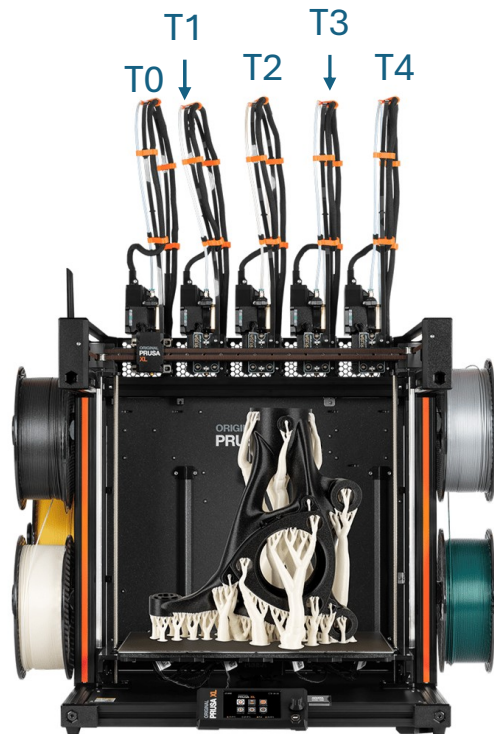
Prusa XL (5-Toolhead)



Prusa CoreOne MMU



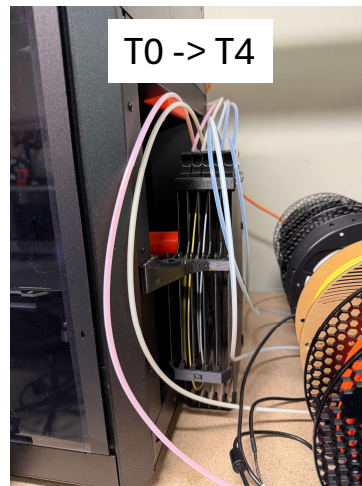
Equipment Overview – Prusa XL



- Spool locations do not always correspond to specific toolheads
- View which colors are associated with each tool by observing the guide tubes

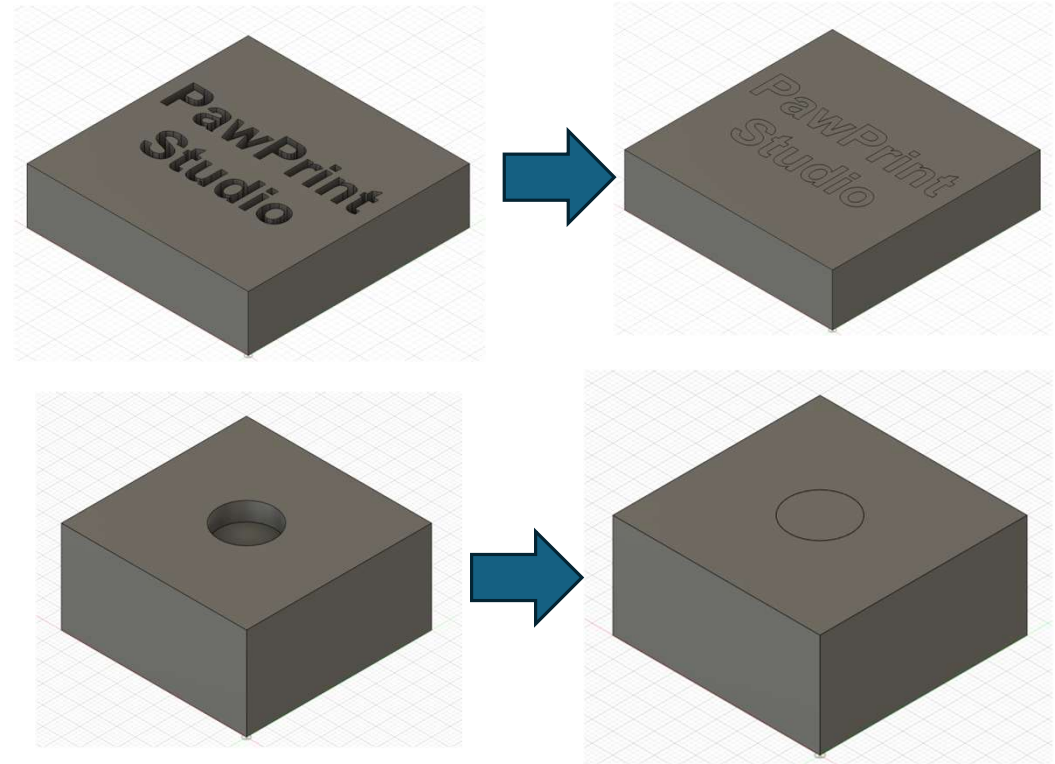
Equipment Overview – Prusa CoreOne MMU

- Spool holders are labeled with the appropriate tool number
- Tool numbers increase from left to right



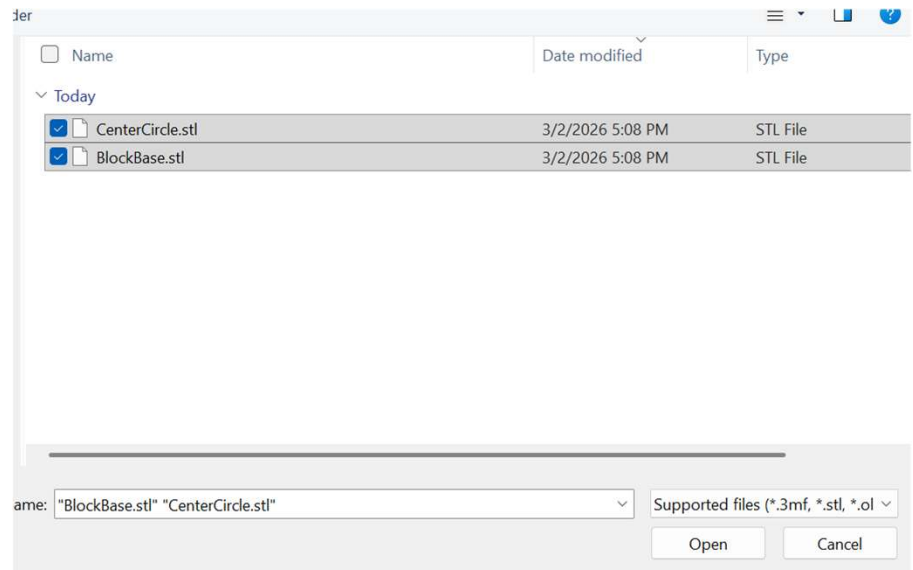
Part Set-Up

- Parts should be in separate bodies/components prior to import in Orca Slicer
- Save both (or all) parts as individual .STL files



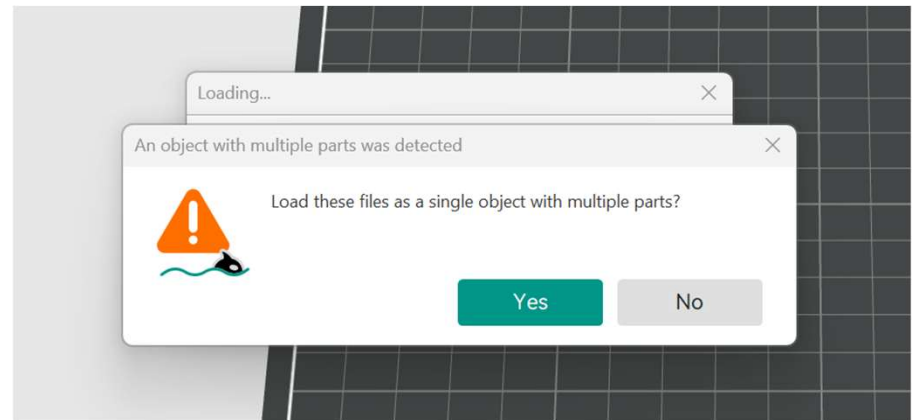
OrcaSlicer Import

- Import both STL files into OrcaSlicer at the same time



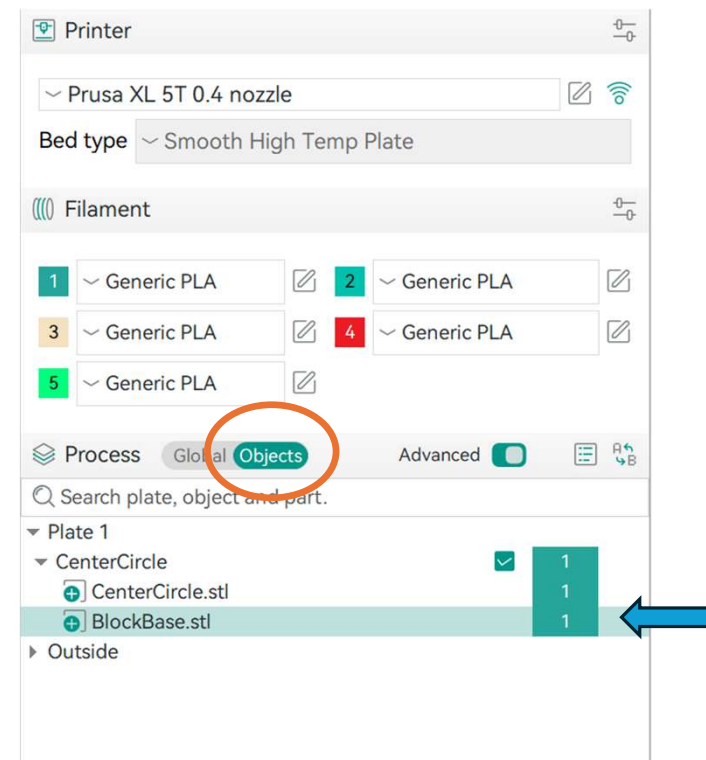
OrcaSlicer Import

- A prompt will appear to treat both bodies as the same object with multiple parts
- Click **YES**



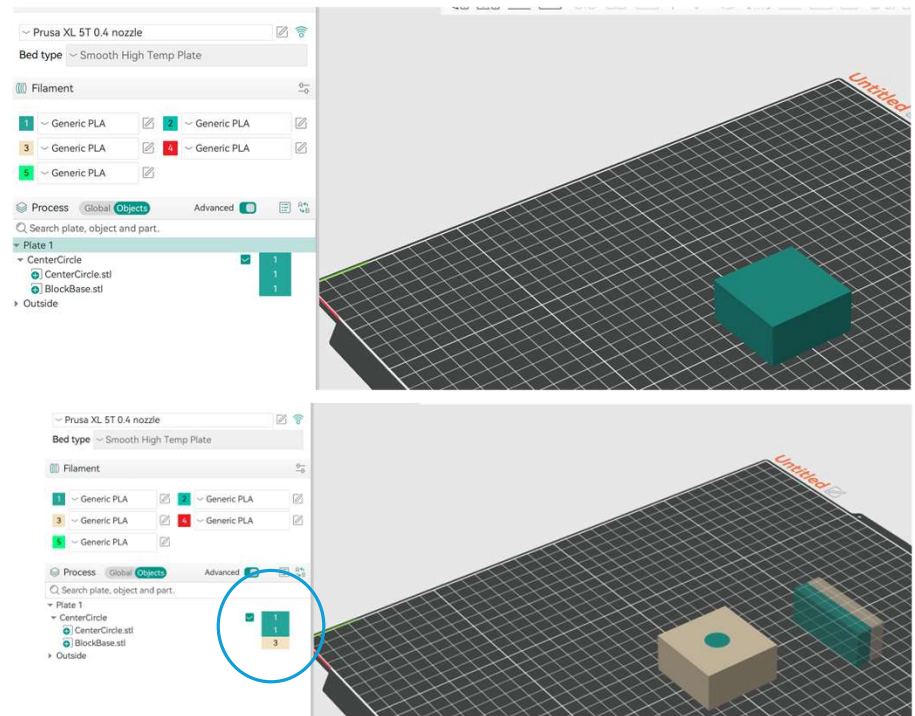
Changing Color

- Upon import, ensure that the appropriate printer is selected
- Select Objects to see your imported STL files
- The color and filament number corresponds to the appropriate toolhead
 - Note that these colors do not always match the loaded colors/materials



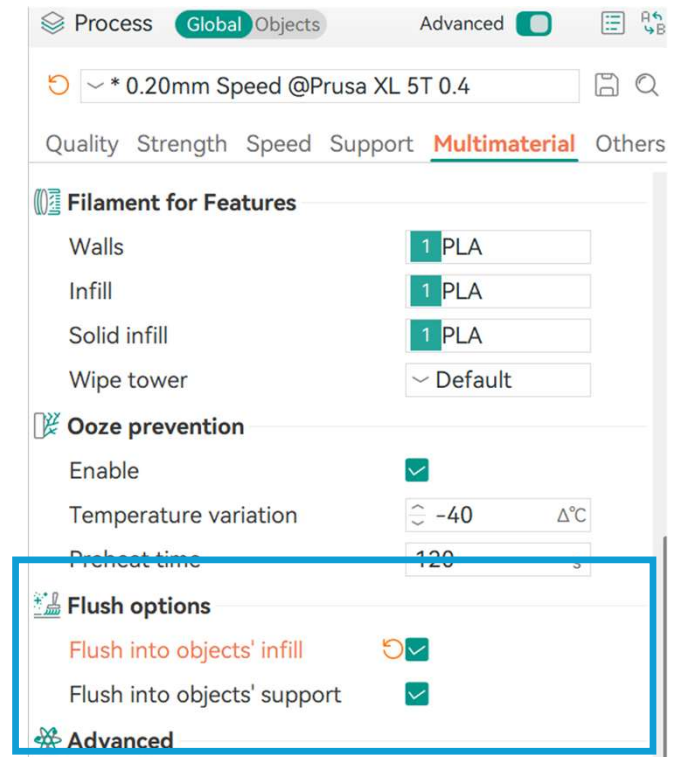
Changing Color

- Change the tool number to match your desired output
- Upon printing with at least two different materials or colors, a wipe tower will appear
 - Also referred to as a prime tower
 - This assist with filament removal from the hot end



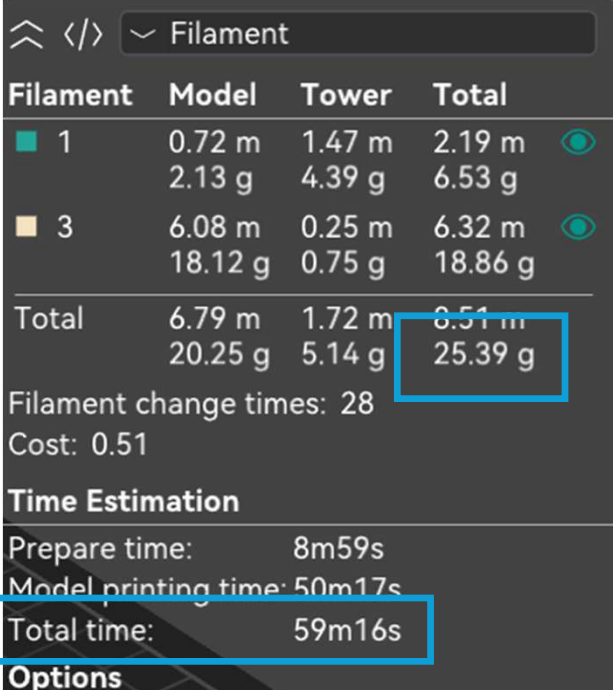
Multimaterial settings

- Flushing into the part infill and/or support will decrease the amount of filament wasted
- You may also adjust specific features of your print based on tool #



Slicer Values

- Note that your total filament consumption will be different than each individual filament
- Your total time estimate will still be the same as in single material prints



The screenshot shows a software interface with a dark theme. At the top, there's a header with a double arrow icon, a code icon, and a dropdown menu labeled 'Filament'. Below this is a table with four columns: 'Filament', 'Model', 'Tower', and 'Total'. The table lists two filament types: '1' (green) and '3' (yellow). Each row shows values for 'Model' and 'Tower' in meters and grams, and a 'Total' in meters and grams. The 'Total' for filament 1 is 2.19 m and 6.53 g, and for filament 3 is 6.32 m and 18.86 g. A summary row at the bottom shows a total of 6.79 m and 20.25 g for the model, and 1.72 m and 5.14 g for the tower, with a combined total of 8.51 m and 25.39 g. This total row is highlighted with a blue box. Below the table, it says 'Filament change times: 28' and 'Cost: 0.51'. A section titled 'Time Estimation' follows, with 'Prepare time: 8m59s', 'Model printing time: 50m17s', and 'Total time: 59m16s'. The 'Total time' row is also highlighted with a blue box. At the bottom, there's a section titled 'Options'.

Filament	Model	Tower	Total
1	0.72 m 2.13 g	1.47 m 4.39 g	2.19 m 6.53 g
3	6.08 m 18.12 g	0.25 m 0.75 g	6.32 m 18.86 g
Total	6.79 m 20.25 g	1.72 m 5.14 g	8.51 m 25.39 g

Filament change times: 28
Cost: 0.51

Time Estimation

Prepare time: 8m59s
Model printing time: 50m17s
Total time: 59m16s

Options

FOM

- Proceed to reserve either a Prusa XL or Prusa CoreOne MMU Printer in FOM
- Refer to the CoreOne and Mini SOP for procedures to do so

Reserving Equipment Time

The screenshot shows a 'New reservation' form with the following fields and annotations:

- Resource:** Prusa CoreOne #1
- User:** You (with a red arrow pointing to the text: "Select your course, research project, or personal account")
- Fee structure:** Internal
- Access Level:** Day time user
- Financial account:** ProjectDescription (with a red arrow pointing to the text: "Select your course, research project, or personal account")
- Reservation:** 10/9 9:30 - 10/9 15:00 (with a green arrow pointing to the text: "Enter end time (your sliced part estimated time + 45 minutes)")
- Estimated cost:** 3.3
- Usage Comment:** (with a green arrow pointing to the text: "Enter starting time")
- Consumables:** PLA Filament 100 gram (with a purple arrow pointing to the text: "Enter estimated filament used from slicer")
- Buttons:** Reserve (circled in orange) and Close
- Footer:** Revision 1.0 (Spring 2026)

Reserving Equipment Time

- Printers are only powered on when equipment is checked out
 - Your reservation is cancelled if you do not check out the equipment within 30 minutes of your reservation time.
- Printers are powered off when equipment is checked back in
 - Manual log-off should occur after your print is finished, however:
 - Auto-log off (and power-off) occurs four hours after reserved end-time
 - If your print extends past the reserved end-time, the next reserved user may log you off, effectively ending your print
- **It is important to set your reservation time correctly. You may extend your reservation time as needed if no reservations are after you.**

Reserving Equipment Time

- Note that print time estimates may vary more compared to single-material prints due to toolhead changes, pre-heating, and prime procedures.
- **After monitoring the initial layers, it is recommended to continuously monitor your print via web cam to adjust your reservation time as needed.**

Print Finished

- When your print is finished, remove your part from the build plate
 - If necessary, use a step stool to reach the build plate or ask an MST for assistance
- **Do not use any tools on the print bed.**
- You may slightly bend the build plate to decrease part adhesion to remove your part. **Do not excessively bend the build plate.**



Print Finished

- Return to the FOM page
- Select your reservation
- In the pop-up, adjust the filament used if different from slicer estimates
 - If using a BYOF printer this should not appear
- Check back in your printer by clicking the **Logoff** Button

Resource is being used

Resource: Prusa CoreOne #1
User: [REDACTED]
Fee structure: Internal
Access Level: Day time user
Financial account: ProjectDescription
Reservation: 10/9 9:30 - 10/9 15:00 (Select a later time to extend your session)
Actual usage: 10/9 9:54 - 10/9 10:07
Resource status report: ☒ OK ☐ Something wrong (details in comments)
Usage Comment:
//Consumables when reserve: PLA Filament 100.00 gram at 0.02/gram// //Consumables at logon: PLA Filament 100.00 gram at 0.02/gram//
Consumables: PLA Filament 100 gram
Record ID 14 created at 10/9 9:00, last updated at 10/9 9:54
Logoff Extend Close

Adjust as needed